

Scientific Communication

Level 1 - Research Foundations | Student Learning Pack

How to use this pack

Read the teaching sections first, complete the worked example, do the short practice task, then take the basic knowledge check in RSRE. External references are optional further learning; you should be able to answer the core assessment from the Academy material.

Lesson 1: Why research must be communicated

Research only becomes useful when other people can understand and evaluate it. Scientific communication includes protocols, posters, oral presentations, reports, manuscripts, journal articles, policy briefs and public explanations. Each format has a different audience and purpose.

A manuscript is a structured scientific report. A journal is the publication venue. Peer review is a quality-assessment process; it does not guarantee that every finding is correct.

Lesson 2: Match message to audience

Take one finding and explain it three ways: in one sentence to another medical student, in a short paragraph to a hospital manager, and in plain language to a member of the public. Keep the meaning and uncertainty consistent. This is a core research skill, not just a writing exercise.

Lesson 3: The basic scientific story

Many health research reports use a logical flow: why the problem matters, what was asked, how it was studied, what was found, what the findings mean, and what remains uncertain. A good paper does not hide limitations or inflate significance. Clear communication follows the evidence rather than trying to make the study sound more impressive.

Lesson 4: Worked example

Finding: "In this sample, 52% of students reported consistent insecticide-treated-net use." A scientific statement can describe the proportion and population. It should not automatically say the intervention caused the behavior unless the design supports that conclusion. Practice rewriting one result without adding a causal claim.

Lesson 5: Check your understanding

Before assessment, make sure you can distinguish a research report, journal, reviewer, author and reader, and explain why communicating limitations is part of scientific quality.

Competency check

After completing this module, you should be able to explain the core idea in your own words, identify a correct and incorrect application, and make a basic research decision using the concept.

RSRE note: The Academy is optional. Experienced researchers may use the diagnostic/competency pathway and enter other RSRE pillars directly.